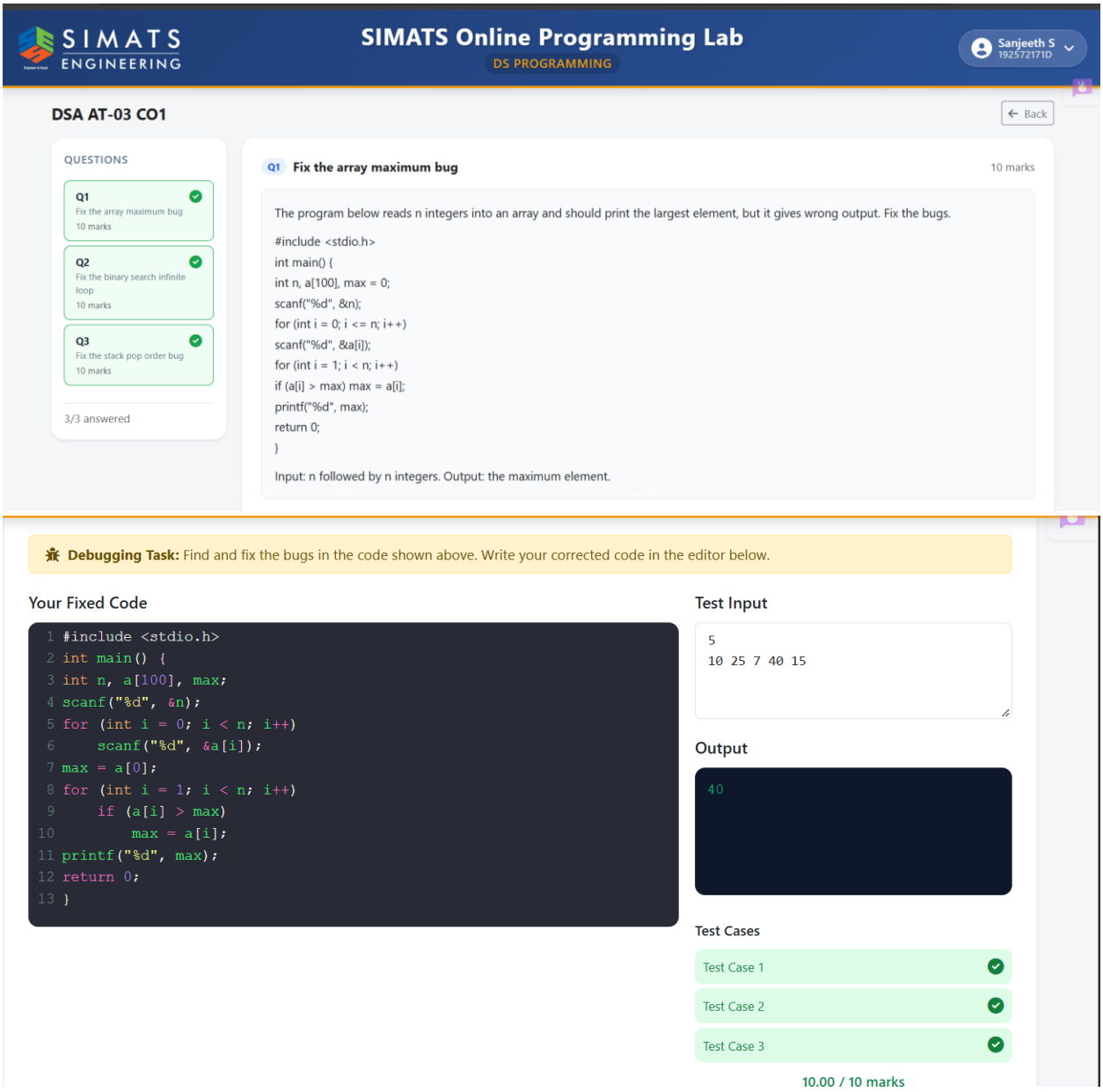
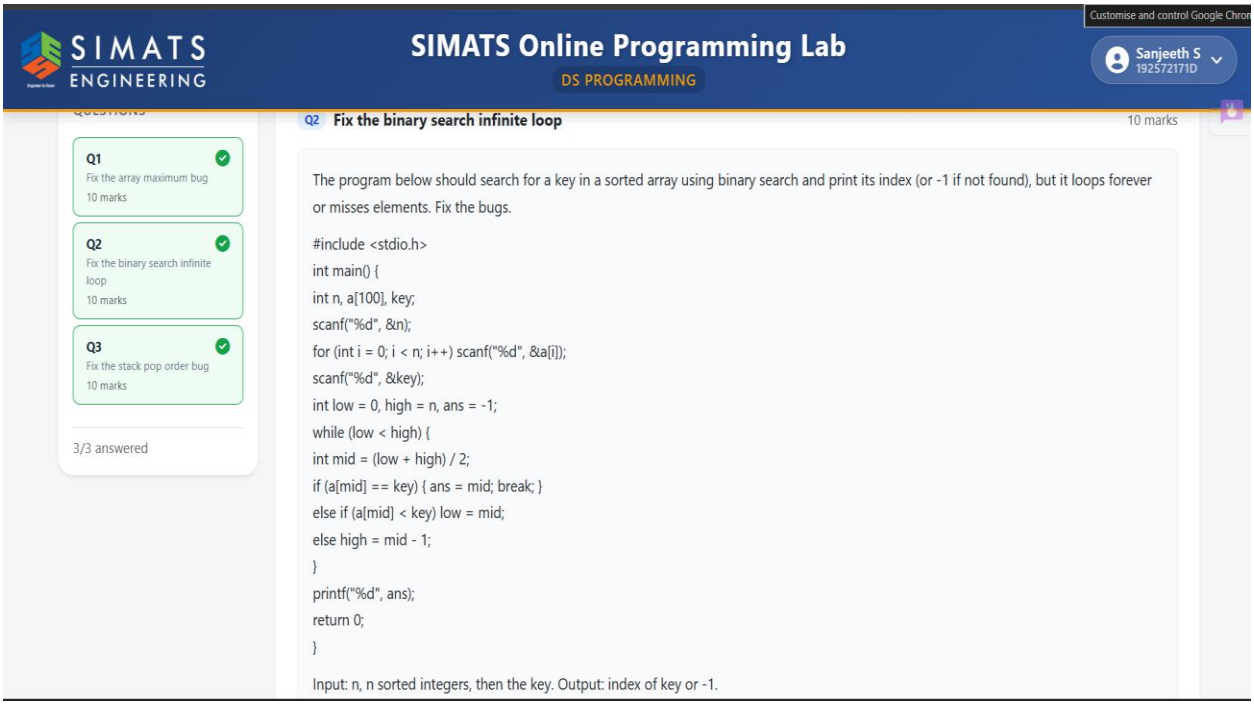
CSA 0307-DATA STRUCTURE

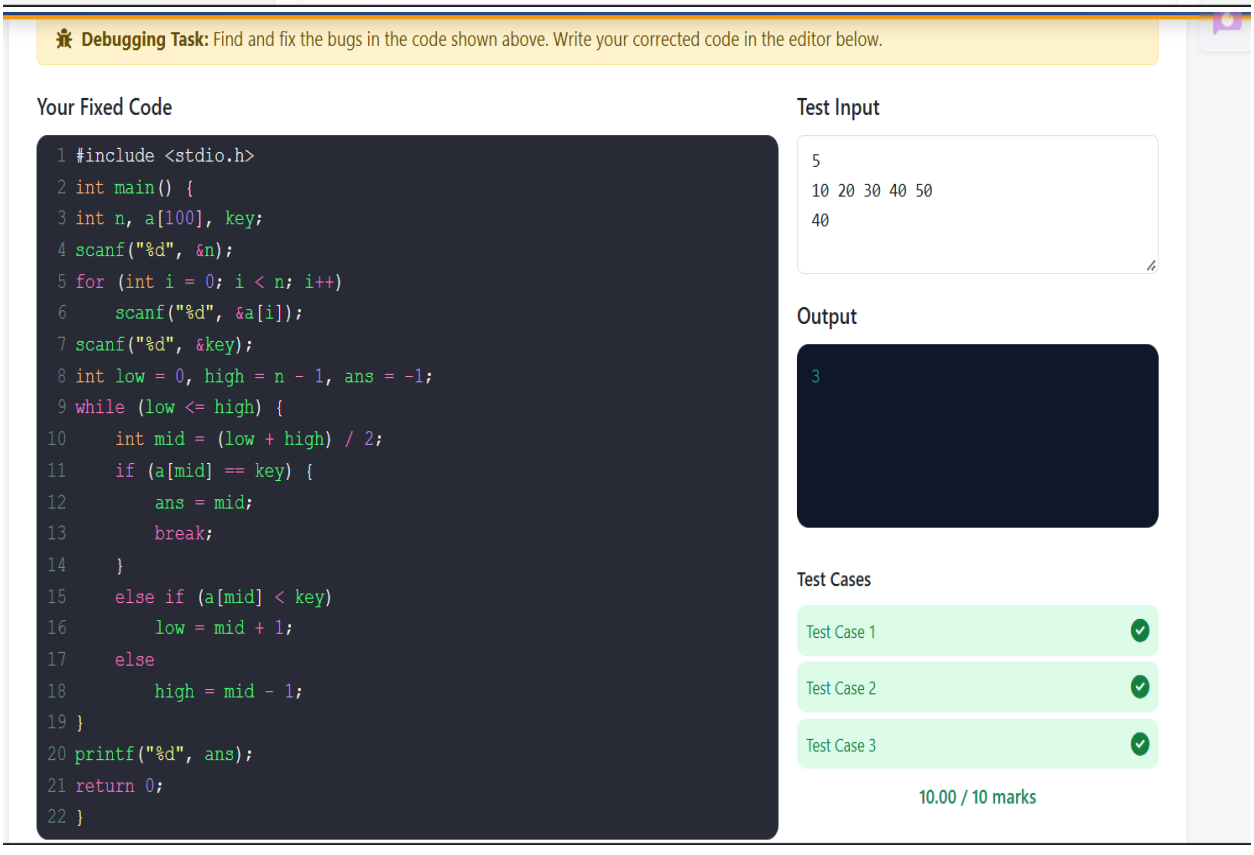
CO1-AT3

1)

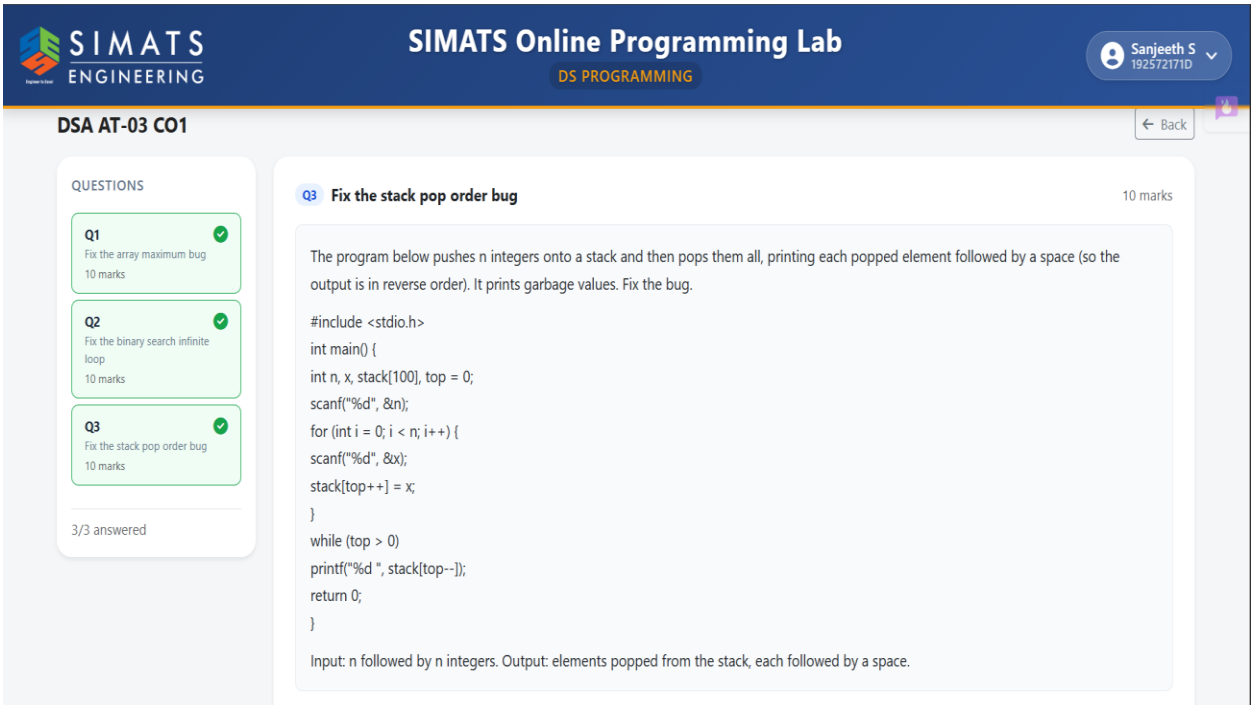


2)


SIMATS ENGINEERING
SIMATS Online Programming Lab
DS PROGRAMMING
Customise and control Google Chrom
Sanjeeth S
192572171D
Q1
Fix the array maximum bug
10 marks
Q2
Fix the binary search infinite loop
10 marks
Q3
Fix the stack pop order bug
10 marks
3/3 answered
Q2 Fix the binary search infinite loop
10 marks
The program below should search for a key in a sorted array using binary search and print its index (or -1 if not found), but it loops forever or misses elements. Fix the bugs.
#include <stdio.h>
int main() {
int n, a[100], key;
scanf("%d", &n);
for (int i = 0; i < n; i++) scanf("%d", &a[i]);
scanf("%d", &key);
int low = 0, high = n, ans = -1;
while (low < high) {
int mid = (low + high) / 2;
if (a[mid] == key) { ans = mid; break; }
else if (a[mid] < key) low = mid;
else high = mid - 1;
}
printf("%d", ans);
return 0;
}
Input: n, n sorted integers, then the key. Output: index of key or -1.


Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
1 #include <stdio.h>
2 int main() {
3 int n, a[100], key;
4 scanf("%d", &n);
5 for (int i = 0; i < n; i++)
6 scanf("%d", &a[i]);
7 scanf("%d", &key);
8 int low = 0, high = n - 1, ans = -1;
9 while (low <= high) {
10 int mid = (low + high) / 2;
11 if (a[mid] == key) {
12 ans = mid;
13 break;
14 }
15 else if (a[mid] < key)
16 low = mid + 1;
17 else
18 high = mid - 1;
19 }
20 printf("%d", ans);
21 return 0;
22 }
Test Input
5
10 20 30 40 50
40
Output
3
Test Cases
Test Case 1
Test Case 2
Test Case 3
10.00 / 10 marks

3)



Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
int main() {
int n, x, stack[100], top = 0;
scanf("%d", &n);
for (int i = 0; i < n; i++) {
    scanf("%d", &x);
    stack[top++] = x;
}
while (top > 0)
    printf("%d ", stack[--top]);
return 0;
}
```

Test Input

```
5
10 20 30 40 50
```

Output

```
50 40 30 20 10
```

Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

10.00 / 10 marks